

Module 0 — Quiz 0 (Start Here Check)

Question bank · open notes · unlimited attempts

Module 0, Quiz 0: Start Here Check

Canvas: Module 0 starter quiz

Use with: [Module 0 Read 0](#), [Syllabus](#), [Module 0 demo lab](#), [Module 0 exercise lab](#)

This quick bank checks startup readiness before Module 1 (aligned with Read 0):

1. Definition of statistics
 2. Five-step statistical process (ask → plan → collect → analyze → conclude)
 3. Three worlds (observe, imagine, computers)
 4. Colab notebook: text, code, and output
 5. Basic R syntax from the Module 0 lab
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Practice prompts

1. What is a good working definition of **statistics** for this course (Read 0)?
 2. How many steps are in the statistical process?
 3. What is the first step? What makes a question “well-defined”?
 4. After the question, put these in order: make a plan, collect data, analyze data, draw conclusions.
 5. What is the purpose of Step 2 (make a plan)?
 6. Give an example from the **world we observe** vs the **world we imagine**.
 7. What roles do **text**, **code**, and **output** cells play in a Colab notebook?
 8. In Read 0, `mean(c(7, 2, 3))` appears in which kind of cell?
 9. Which line assigns 10 to `x` using course style (`<-`)?
 10. Which code creates `v <- c(1, 2, 3, 4)`?
 11. What does `length(v)` return for `v <- c(1,2,3,4)`?
 12. In the Module 0 baby-matching simulation, are draws with or without replacement within a trial?
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Answer key

1. Statistics is the science of **using data to answer questions and make decisions**.
2. Five steps.
3. **Ask a well-defined question** — specific, measurable, and answerable with data.
4. Make a plan → collect data → analyze data → draw conclusions.
5. Decide what to collect, how to analyze, and how to conclude **before** data are collected.
6. Observe: sleep you logged this week. Imagine: mean muffin size in 2026 vs 2006 (model/population idea).
7. Text → people; code → instructions for the computer; output → what the computer produces.
8. A **code** cell (R instructions).
9. `x <- 10`.
10. `v <- c(1, 2, 3, 4)`.
11. 4.

12. Without replacement.